Satish Chandra Pamulapati

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Profile

LLM and data-oriented ML engineer with hands-on experience building Retrieval-Augmented Generation (RAG) workflows, report and feedback summarization pipelines, and text-to-SQL proof-of-concepts. Comfortable working from messy inputs to something usable: data preparation, prompt/logic iteration, evaluation, and clear outputs for stakeholders. Strong Python And Algorithmic foundation with practical experience in production-style prototyping.

Education

MSc, Machine Learning Sep 2025 - Present
Lund University, Sweden

- Hosted and conducted a Computer Vision workshop for ML students (fundamentals, practical pipelines, real-world applications).
- Working on Multi-kilohertz tracking of insect wings using event-based cameras and sensor fusion

BTech, Electronics and Communication Engineering
SRM Institute of Science and Technology, India

Technical Skills

Programming: Python, NumPy, Pandas, Linux, Git,
LLM / NLP: SQL, Text Summarization, RAG, Text-to-SQL, Prompt Engineering
ML: Supervised Learning, Unsupervised Learning, Feature Engineering, Hyperparameter Tuning
Frameworks / Platforms: PyTorch, TensorFlow, scikit-learn, OpenCV, Hugging Face, Google Cloud Platform

Experience

Machine Learning Specialist (Contract), Kolomolo AB (Sweden) June 2026 - Present

- Working on Internal Computer Vision and Machine Learning Projects

Associate Data Analyst / Machine Learning Intern, Zelar Sep 2024 - Aug 2025

- Built end-to-end RAG workflows using LLMs to support document understanding and report summarization
- Delivered multiple ML proof-of-concepts that helped convert **2 client pilots into long-term production contracts**, supported by clear demos and result summaries.
- Developed image restoration steps (denoising and artifact reduction) to improve degraded scans before OCR.
- Implemented automated image redaction workflows for sensitive information as part of document processing pipelines.
- Prototyped a 3D pose classification pipeline for abnormal / atypical movement detection using computer vision.

Projects

Text-to-SQL Prototype (LLM-assisted Querying) (Work experience at Zelar)

- Built an LLM-based text-to-SQL proof-of-concept to translate natural language requests into SQL queries with iterative prompt and logic refinement.
- Focused on making results usable for non-technical users (clear outputs, simple interaction flow, and reliable query generation behavior).

Report and Feedback Summarization (Work experience at Zelar)

- Implemented summarization pipelines to turn long reports and user feedback into concise, structured summaries for faster decision-making.
- Iterated on prompt structure and evaluation to reduce noise and improve consistency across varied inputs.

AI Review Summarizer and Priority Dashboard (AI Product Engineer) (Work Experience at Zelar)

- Built an LLM-based review analysis pipeline to turn high-volume user feedback into structured insights for non-technical stakeholders.
- Designed a 4-level prioritization scheme (P0 to P3) to surface critical bugs, recurring issues, and feature requests based on urgency and impact.
- Generated dashboard-ready outputs (issue category, short summary, representative examples, and recommended next action) to support manager planning.
- Focused on consistency and usability by iterating prompt templates and output schemas(prompt engineering techniques) for stable prioritization across varied writing styles.

YouTube Video Description Generation (GPT-2.5 API) Aug 2022 - Oct 2022

- Built an NLP application to generate video descriptions from provided tags, focusing on semantic alignment and content relevance.
- Reported **85% accuracy** and validated outputs using structured evaluation on a tag-to-text dataset.

Publications (IEEE Xplore)

- An Intelligent Computer Vision based Application for Student Monitoring in Classroom
- Deep Learning-based Segmentation and Computer Vision-based Ultrasound Imagery Techniques

Training / Internships

Amazon ML Summer School Jul 2022

- Completed an intensive ML curriculum covering dimensionality reduction, probabilistic models, deep neural networks, and reinforcement learning, with hands-on experimentation.

Machine Learning Intern, Feynn Labs May 2022 - Jul 2022

- Developed a Python-based chatbot (**85% accuracy**) using NLP pipelines including intent classification and dialogue management.

Certifications

Meta Database Engineer (Coursera)
Stanford Algorithms Specialization
Harvard edX: Data Science and Machine Learning